

SUBJECT: Data Structures and Algorithms

1. Introduction

A linked list is a linear data structure, in which the elements are not stored at contiguous memory locations. The elements in a linked list are linked using pointers.

2. Node Structure

A linked list consists of nodes where each node contains a data field and a reference(link) to the next node in the list.

```
class Node {  
    int data;  
    Node next;  
}
```

3. Completing Insertion

To insert a node at the beginning, time complexity is $O(1)$. To insert at the end, it takes $O(N)$. Insertion is faster than arrays because no shifting is required.

4. Example Code

```
function insertAtBeginning(head, data)  
{  
    let newNode = new Node(data);  
    newNode.next = head;  
    return newNode;  
}
```